

Detecting Anomalous Corporate Expense Transactions

A machine learning case study · Siddhartha Baniya

Project Overview

Traditional rule-based corporate expense systems often miss subtle or emerging fraud patterns. This project explores how supervised machine learning can enhance fraud detection in finance workflows — reducing manual audit burden while surfacing transactions that warrant deeper review.

Dataset

- Source: Kaggle — Fraud Detection Dataset (PaySim-style transactional data)
- Records: 6,362,620 transactions
- Attributes: 11 features including transaction type, amount, sender and receiver balances before and after the transaction
- Target: isFraud — binary classification
- Fraudulent transactions concentrated in CASH_OUT and TRANSFER types with anomalous balance behavior

Models & Results

Two supervised classification models were trained and compared:

- Decision Tree — baseline interpretable model
- Random Forest — ensemble model selected for production-style evaluation

Comparative performance:

Model	Accuracy	Precision	Recall	AUC
Decision Tree	99.33%	100%	33.33%	0.661
Random Forest	99.88%	99.91%	50.28%	0.782

Random Forest outperformed Decision Tree on AUC and recall, making it the better fit for fraud detection where missing a true fraud (false negative) carries higher cost than flagging a legitimate transaction (false positive).

Tools

- RapidMiner — modeling and pipeline orchestration
- Kaggle dataset — public training data

Why It Matters

- Surfaces hidden or emerging fraud patterns that rule-based systems miss

- Reduces manual audit overhead by prioritizing the highest-risk transactions for human review
- Improves transparency and trust in financial workflows
- Scalable approach applicable to corporate expense platforms (e.g. Navan, Concur, Ramp)

Honest Limitations & Next Steps

Random Forest's recall of 50% means roughly half of true fraud cases are missed. For production deployment this is the binding constraint — the next iterations would prioritize improving recall through:

- Class-imbalance handling (SMOTE, class weights, anomaly-detection framing)
- Unsupervised exploration: Isolation Forest and Autoencoders to catch patterns the supervised model has not seen
- NLP on expense narratives and merchant descriptions for richer features
- Real-time scoring with monitored thresholds and feedback loops from auditor decisions